

6.3.1

脉宽

$$t_w = 1.1 RC$$

当 $R_{\text{电}} = 0$ 时 $t_{w1} = 1.1 \times 10^4 \times 10 \times 10^{-6} = 0.11s$

$R_{\text{电}} = 100k\Omega$ 时 $t_{w2} = 1.21s$

$\therefore$ 范围为 $0.11s \sim 1.21s$

6.3.2

回差电压 $\Delta U_T = \frac{1}{2} U_{CO} = 2V$

6.3.3

(1). S 闭合时 $U_C(0^+) = U_C(0^-) = 0$ $U_R(0^+) = U_{DD}$

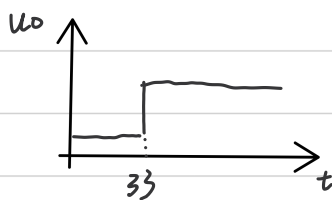
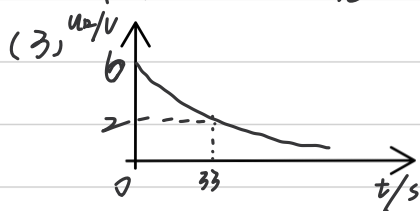
LED₁ 发光 LED₂ 不发光

当 $U_R < \frac{1}{3} U_{DD}$ 时 LED₁ 灭 LED₂ 发光

$$U_R(t) = U_R(\infty) + [U_R(0^+) - U_R(\infty)] e^{-\frac{t}{\tau}} = U_{DD} e^{-\frac{t}{\tau}}$$

当 $U_R(t) = \frac{1}{3} U_{DD}$ 时 $U_{DD} e^{-\frac{t}{\tau}} = \frac{1}{3} U_{DD} \quad \therefore t = 3\tau$

(2) 稳态后. LED₂ 亮 LED₁ 灭



8.1.1

(1) $P_{0max} = \frac{1}{2} \frac{U_{CC}^2}{R_L} = 14.06W$

(2) $P_0 = \frac{1}{2} \frac{1}{R_L} (U_{CC} - U_{CES})^2 = 12.25W$ $\eta = \frac{\pi}{4} \frac{U_{CC} - U_{CES}}{U_{CC}} = 73.3\%$

(3) $U_i = 5V$ 时. $U_O \approx U_i$ $P_0 = \frac{U_O^2}{R_L} = 3.125W$

8.1.2

(1) $A_{uf} \approx 1 + \frac{R_2}{R_1} = 1 + \frac{39}{3} = 14$

(2) $P_{0max} = \frac{1}{2} \frac{U_{om}^2}{R_L} = 9W$

